

Distributed Systems Design Specification

Project: SecondSelf High-Availability Knowledge Core

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1. Architectural Overview

The SecondSelf system is designed as a hybrid distributed architecture balancing local-first data privacy with cloud-scale inference. The system implements a three-tier design:

- * Ingestion & Raw Stash Layer: High-throughput async file receiver storing immutable raw payloads.
- * Knowledge Graph & Indexing Tier: Dense vector storage (FAISS 384-d) combined with SQLite FTS5 BM25.
- * Inference & Hybrid RAG Engine: Groq LPU fast token streaming with fallback to Gemini 1.5 Pro.

2. Consistency & CAP Theorem Strategy

In accordance with PACELC principles, SecondSelf prioritizes High Availability and Partition Tolerance (AP) for ingestion, while maintaining Strong Consistency (CP) for document versioning and deduplication:

- * Data Partitioning: Notes and binary assets partitioned by User UUID with deterministic hash sharding.
- * Vector Index Synchronization: Inverted file index (IVFFlat) rebuilds incrementally with dirty-bit tracking.
- * Conflict Resolution: Last-Write-Wins (LWW) utilizing monotonic logical timestamps.

3. Multi-Tier Caching Strategy

- * L1 Memory Cache: Python LRU in-process cache for frequently accessed graph neighbor queries (TTL: 60s).
- * L2 Distributed Cache: Redis cluster for token usage budgets, session states, and rate limiting counters.
- * L3 Vector Buffer: Memory-mapped FAISS index files for zero-copy similarity lookups.

Distributed Systems Design Spec (Continued)

4. Rate Limiting and Token Bucket Implementation

To safeguard external LLM providers (Groq LPUs and Google Gemini), all inbound user requests pass through a Token Bucket rate limiter:

- * Capacity: 60 requests per minute per IP address.
- * Refill Rate: 1 token per second.
- * Burst Allowance: Up to 15 concurrent requests before HTTP 429 Too Many Requests is returned.

5. Failover and Disaster Recovery Matrix

- * Primary Database: SQLite in Write-Ahead-Logging (WAL) mode with automated hourly volume snapshots.
- * Cold Backup: S3/MinIO daily encrypted bundle upload.
- * Recovery Time Objective (RTO): < 5 minutes.
- * Recovery Point Objective (RPO): < 1 hour.

6. Observability and Health Checks

Endpoints /health, /health/db, /health/vectors provide sub-10ms telemetry on active database connections, vector memory usage, and background worker queue depth.